

ELI — Capability Catalogue

ELI v2.4.81

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ELI Capability Catalogue — every action & module, what it actually does

Purpose. A systematic, ground-truth catalogue built by reading the real handlers and modules — not summarised from memory. It exists because conversational summaries of a 193,503-line project (eli/, measured 2026-09-25) keep undershooting; this is the persisted, exhaustive map.

Method. Action list comes from the live `capability_manifest.json` (**228** entries; 187 routable, 206 in the executor’s supported list, 211 in either), verified against the `executor_enhanced.py` dispatch. **The always-current, auto-generated action list with activation phrases is `capabilities_and_actions.md`** — this catalogue is the deeper module-level read. Behaviour

grounded in the handlers and the subsystem blueprints. Where a description is inferred from name+structure rather than a line-by-line handler read, it is marked (*inferred*).

Coverage: every action in the manifest (Part 1), every module under eli/ (Parts 2-4 and 7).

Headline finding: 228 is real but aliased

The manifest's 228 entries are honest (*measured* by capability_sync, not asserted) but inflated by **alias families** — multiple action names routing to one behaviour. Collapsed, there are roughly **~110 distinct capabilities**. Alias families are grouped below so the real surface is visible.

1. Conversation & reasoning

Action(s)	What it does
CHAT	The default — full cognitive pipeline (router → gradient orchestrator → mode algo → governed output).
ANSWER, DIRECT_RESPONSE, SAY	Direct/short response surfaces (lighter than CHAT). (<i>inferred for some</i>)
SET_AI_MODE	Set the reasoning mode (Quick / Normal / Advanced / Research / Expert). Legacy names still accepted as input.
SEQUENCE	Multi-step action chaining (run a sequence of actions).
NOOP	No-op.

2. OS & application control

Action(s)	What it does
OPEN_APP = OPEN_APPLICATION = LAUNCH_APP	Launch an app, resolved via the system_index (your machine's executables — thousands, indexed live) + portable_app_control. On failure → remediation offer (real apt/snap/flatpak install on confirmation).
CLOSE_APP = CLOSE_APPLICATION = EXIT_APP = QUIT_APP	Close/quit an app.
FOCUS_APP, HIDE_APP	Focus / hide a window.
MINIMISE_APP = MINIMIZE_APP, MINIMISE_WINDOW = MINIMIZE_WINDOW, MAXIMISE_WINDOW	Window state control (UK/US spellings aliased).
MINIMISE_ALL, RESTORE_WINDOWS, TILE_WINDOWS, NEXT_WINDOW, PREVIOUS_WINDOW	Desktop window management.
SWITCH_WORKSPACE	Switch virtual desktop/workspace.
OPEN_SYSTEM_SETTINGS, OPEN_AUDIO_SETTINGS, OPEN_POWER_SETTINGS, OPEN_FILE_SYSTEM, OPEN_COMMUNICATION_HUB, OPEN_MEDIA_HUB, OPEN_NETWORK_BROWSER	Open specific OS setting panels / hub launchers.
OPEN_IDE, OPEN_IN_IDE	Open the IDE (optionally at a file).
OPEN_BROWSER, OPEN_URL	Open browser / a URL (local hand-off; respects net toggle for fetches).

3. Input & screen control

Action(s)	What it does
KEYBOARD	Send keypresses (<code>os_controller.press_key</code>).
MOUSE_CONTROL	Move/click the mouse (<code>os_controller.mouse_click</code>).
VOLUME	Get/set system volume.
SET_CLIPBOARD, GET_CLIPBOARD	Read/write the clipboard.
SCREENSHOT	Capture the screen.

4. Gaze (webcam eye-tracking)

Action(s)	What it does
GAZE_ENABLE, GAZE_DISABLE	Start/stop the MediaPipe gaze engine (writes <code>latest_gaze.json</code> @10Hz).
GAZE_CALIBRATE	Run gaze calibration.
GAZE_CLICK	Move the cursor to where you're looking and click.
GAZE_STATUS	Report gaze-engine state.

5. Media

Action(s)	What it does
PLAY_MEDIA	Play a song/playlist on a named platform (Spotify playlist-tab / YouTube Mix-radio); honest about reachability; explicit platform never falls through to a second.
PAUSE_MEDIA, STOP_MEDIA, NEXT_MEDIA, PREVIOUS_MEDIA, REPEAT_MEDIA, SHUFFLE_MEDIA	MPRIS/playerctl transport control.
MEDIA_CONTROL	Generic media-control dispatch.
SKIP_YOUTUBE_AD	Skip a YouTube ad. (<i>inferred</i>)

6. Files & documents

Action(s)	What it does
CREATE_FILE, CREATE_FOLDER, READ_FILE, LIST_DIR	Filesystem ops (path-allowlist gated).
SUMMARIZE_FILE	Summarise a file's contents (also the route for "read your persona.auto.txt / settings").
CONVERT_DOCUMENT	Convert document formats.
CREATE_DOCUMENT = CREATE_DOC = DOC_GENERATE = GENERATE_DOCUMENT = WRITE_DOCUMENT	Generate a document (the Report-Builder family).
ANALYZE_CSV, ANALYZE_PDF, ANALYZE_PDF_FOLDER, ANALYZE_IMAGE, OCR_IMAGE	Local file-type analysers (CSV stats, PDF text/structure, image VL description, OCR).

7. Vision & screen understanding

Action(s)	What it does
ANALYZE_IMAGE	Local GGUF vision-language description of an image.
OCR_IMAGE	Tesseract OCR text extraction.
AMBIENT_VISION	Toggle periodic background screen glances.
SCREEN_LOCATE	OCR-locate a named UI element on screen (“the button that says X”).
SCREEN_READ_ANALYZE	Screenshot → analyse what’s on screen.
IMAGE_STATUS	Report vision/image-engine state.

8. Coding & code-repair

Action(s)	What it does
CODE_SOLVE	The frontier coding agent: plan → DAG decompose → UCB tree search → verify (syntax/exec/tests) → repair → bug-memory.
GENERATE_SCRIPT = CREATE_SCRIPT = WRITE_SCRIPT = GENERATE_CODE = WRITE_CODE	Generate a script — routes through the coding agent (falls back to inline gen).
GENERATE_PROJECT	Generate a multi-file project.
EXAMINE_CODE	Tiered file scan (syntax/import → lint → gated LLM logic review) → offer fix.
FIX_FILE, CONFIRM_CODE_FIX, CANCEL_CODE_FIX	Apply/confirm/cancel a verified auto-reverting code fix.
FILE_AUDIT, SHOW_DIFF, CODE_CHANGES	Audit a file / show a diff / report recent code changes.

9. Self-management & self-improvement

Action(s)	What it does
SELF_REPORT	Grounded runtime self-report (deterministic, from live state).
SELF_ANALYZE, SELF_IMPROVE, SELF_IMPROVEMENT_LOG	Analyse failures / surface improvement proposals / show the improvement log.
SELF_PATCH	Generate+apply a verified self-patch.
SELF_TEST	Run self-tests.
SELF_UPGRADE = SELF_UPDATE	Maintenance: git pull, pip, rebuild FAISS/KG, refresh manifest + system index.

10. Introspection & audits (grounded, deterministic)

Action(s)	What it does
RUNTIME_STATUS, RUNTIME_AUDIT	Live runtime report; RUNTIME_AUDIT also runs health probes (plugin mgr, memory, agent bus, habit integrity, recent failures).
GUI_RUNTIME_AUDIT, IMPORT_AUDIT, DIAGNOSE_WRAPPERS, RESOLVE_RUNTIME_PATHS	GUI/runtime audits, import smoke-audit, executor-wrapper diagnostic, path resolution.
FRONTIER_STATUS, ELI_IDENTITY_AUDIT	Full cross-system matrix / identity-classification audit.

Action(s)	What it does
COGNITION_STATUS, EXPLAIN_COGNITION_RUNTIME, EXPLAIN_MEMORY_RUNTIME EXPLAIN_ALL_REASONING_MODES, REASONING_MODE_STATUS EXPLAIN_LAST_RESPONSE, EXPLAIN_LAST_FAILURE, NAME_SOURCE_AUDIT, ROUTING_FAULT_EXPLAIN HARDWARE_PROFILE, GPU_STATUS, GET_STATUS, AWARENESS_STATUS HELP, LIST_CAPABILITIES	Cognition + memory runtime explainers (verbatim, deterministic). Describe the 5 reasoning modes / current mode. Explain the last answer/failure, the source of your name, or a routing fault. Hardware/GPU/general/awareness status. Help / list the capability surface.

11. Memory & identity

Action(s)	What it does
MEMORY_RECALL, MEMORY_STORE, MEMORY_STATS, MEMORY_STATUS PERSONAL_MEMORY_SUMMARY, PERSONAL_MEMORY_DEEP_EXPLAIN USER_IDENTITY_SUMMARY, USER_INFO_REPORT, REFRESH_USER_INFO SET_USER_NAME MESSAGE_TIME_QUERY CLEAR_CHAT_HISTORY PERSONA_LOCK_SET, PERSONA_LOCK_CLEAR, PERSONA_LOCK_STATUS	Recall/store memories; memory inventory/stats. “What do you know about me” (clean) / deep memory-internals explain. Who-you-are summary / full profile report / rebuild the living profile. Set the user’s name. “What time did I first/last message (today)” — from conversation_turns. Clear chat history. Lock/unlock/inspect the persona.

12. Habits, proactivity & goals

Action(s)	What it does
CONFIRM_HABIT, DECLINE_HABIT, HABIT_STATUS MORNING_REPORT PROACTIVE_START, PROACTIVE_STOP, PROACTIVE_STATUS EXECUTE_GOAL BACKGROUND_JOBS, CHECK_JOB	Approve/decline a detected habit; report habits. The consolidated morning brief (news digest + activity + attention). Control the proactive daemon. Execute a mission-layer goal (governed). List background tasks / check a job id.

13. Self-healing remediation (Linux)

Action(s)	What it does
PREPARE_REMEDIATION CONFIRM_PENDING_REMEDIATION, CANCEL_PENDING_REMEDIATION CHECK_TARGET_STATUS	Diagnose a failure + build a repair plan (e.g. install a missing app via apt/snap/flatpak). Execute / cancel the pending repair (sudo terminal, lock-handling, verification). Check whether a target app/path is now present.

14. Voice & transcription

Action(s)	What it does
DICTATE, TRANSCRIBE	Dictation / transcribe audio (faster-whisper).
LISTEN_FOR_COMMAND	One-shot listen.
STT_DIAGNOSTICS, VOICE_DIAGNOSTICS	Diagnose the STT/voice pipeline (mic, ducking, wake gate).
SAY / SPEAK (<i>plugin: tts</i>)	Speak text (Piper TTS).

15. Time, timers & utility

Action(s)	What it does
TIME = GET_TIME, DATE = GET_DATE	Current time / date.
SET_TIMER, SET_ALARM	Timers/alarms. (<i>inferred</i>)
CHECK_CHRONAL_ALIGNMENT	Easter-egg: “Chronal alignment nominal. Local time: ...” (a playful time report).
DATA_FABRICATOR	Generate a document from a topic (via CREATE_DOCUMENT) and open it in an editor (code/gedit/kate/nano).
RUN_CMD, SHELL_EXEC	Run a shell command — fail-closed (blocked unless ELI_ALLOWED_CMDS is set or ELI Full Control is on).

16. Plugins (management)

Action(s)	What it does
PLUGIN_LIST, PLUGIN_SEARCH, PLUGIN_STATUS	List installed / search registry / status.
PLUGIN_INSTALL, PLUGIN_UNINSTALL, PLUGIN_ENABLE, PLUGIN_DISABLE	Install/remove/enable/disable a plugin.

17. Plugin-backed capabilities

Action	Plugin	What it does
WEB_SEARCH	web	DuckDuckGo/SearXNG web search (toggle-gated).
GET_WEATHER	weather	Local geocode + open-meteo (toggle-gated).
NEW_NOTE/WRITE_NOTE, LIST_NOTES, SEARCH_NOTES	notes	Markdown notes with FTS.
ADD_EVENT, LIST_EVENTS	calendar	ICS calendar events.
POMODORO_START, POMODORO_STOP, POMODORO_STATUS	pomodoro	Focus timers.
CPU_USAGE, RAM_USAGE, SYSTEM_STATS	system_stats	CPU/RAM/disk/network.

Action	Plugin	What it does
SMART_HOME	executor and device server	Smart-home control via ELI's own MQTT device server (ESPHome / Tasmota / Zigbee2MQTT); rooms, scenes, real automations. Home Assistant removed.
SPEAK	tts	Text-to-speech.
NEWS_FETCH	(executor + news tool)	Fetch + synthesise news (rolling 3-hourly reflections → 24h digest).

(Also reachable: *web_automation plugin* — Playwright browser navigate/search/screenshot; *document_reader plugin* — PDF/docx read+index.)

Reading note

Alias families that collapse the count: app-open (3), app-close (4), minimise (4 across window/app + UK/US), document-generate (5), script/code-generate (5), time/date (2 each), self-upgrade/update (2). Removing pure aliases yields **~110 genuinely distinct capabilities** — still an unusually broad surface for a local single-user assistant, spanning OS control, media, files, vision, gaze, voice, coding, memory, introspection, autonomy, remediation, and plugins.

Part 2 — runtime/ module catalogue (95 files, 35.4k lines)

The largest package. It's the **grounding/governance + introspection + plumbing** layer that wraps the probabilistic model. Grouped by function:

Grounding & anti-confabulation

Module	LOC	Role
deterministic_grounding_gate.py	3385	Renders control/status answers directly from live runtime (bypasses the model). One ordered <code>render_action</code> pipeline of layers composed by <code>_stack</code> ; an unhandled action returns a <code>missing_deterministic_renderer</code> surface.
grounding_escalation.py	698	When a checkable factual question is poorly grounded by the bus, escalates through agent tiers instead of letting the model confabulate (the "Eminem's real name" failure class).

Module	LOC	Role
diagnostic_patterns.py	112	Regexes that catch vague/dynamic status confabulation (“currently processing updates...”) and image-status fabrication.
control_contracts.py	1242	Control-action evidence contract: build evidence → validate the model’s output doesn’t violate it → finalise.
persistence_gate.py	204	Gates what gets stored — refuses to persist internal dumps / error-pattern noise as memory.

Self-honesty / introspection / self-reporting

Module	LOC	Role
live_introspection.py	736	Live runtime snapshot, last trace, stored user name, mines user-fact candidates, agents-for-action, build_report.
deterministic_introspection.py	591	The engine’s live diagnostic dispatcher (handle_diagnostic_action) — deterministic answers for RUNTIME_STATUS / EXPLAIN_* / IMPORT_AUDIT.
truth_report.py	436	Runtime truth report (git, nvidia, GGUF runtime, import health).
frontier_status.py	433	Full cross-system status matrix (run-time/memory/awareness/proactive/image/world)
eli_identity_audit.py	416	Identity-classification audit (source inventory, contract counts, capability matrix).
reasoning_status.py	199	Current reasoning-mode reporting.
experimental_inventory.py	146	Inventory of experimental projects.

Self-improvement & code-awareness

Module	LOC	Role
self_improvement.py	1736	Failure logging/clustering, patch generate→verify→apply→ auto-revert , full patch cycle, plugin-stub gen.
code_examiner.py	910	Tiered file error scan (syntax/import → lint → gated LLM) → offer → verified fix.

Module	LOC	Role
code_monitor.py	261	Detects source changes via git diff, classifies by subsystem, summarises for memory/context (ELI is aware of its own code changes).
capability_sync.py	419	AST-discovers the live capability surface, diffs, writes capability_manifest.json — this is why the count is <i>measured</i> , not asserted.
generated_script_guard.py	1084	Validates LLM-generated scripts, quarantines invalid ones , and ships vetted canned scripts for known patterns (GPU-watch, redshift, etc.).

Self-healing remediation

Module	LOC	Role
grounded_remediation.py	1717	Diagnoses failures (missing app/path/browser) → builds an apt/snap/flatpak repair plan → offers → executes (sudo terminal, lock-handling, verify). Writes incident records.
incident_log.py	21	

Awareness & boot

Module	LOC	Role
awareness_boot.py	367	Boots all awareness subsystems at startup, returns an AwarenessState the engine queries.
action_commitment.py	182	Detects when ELI's reply COMMITS to an action (so the pipeline re-runs and actually does it — no fake actions).

Autonomy / operator (governed)

Module	LOC	Role
operator_state.py, operator_feed.py	~177	Operator console state: proposals, goals, self-model status, event feed.
pending_proposal.py	155	Pending-proposal state (extract/set/clear).
approval_engine.py	103	Who may propose / evaluate a proposal record (governance).

Response surfaces & governance

Module	LOC	Role
user_visible_response_surface.py	367	Installs the engine's user-visible response surface (runtime/identity/name-source formatting + streaming coercion).
visible_output.py, visible_text.py	~150	Central visible-output contract; stringify/sanitise streamed output.
response_policy.py, response_contracts.py	~160	Classify response mode; per-action contracts.
final_response_provider.py	~85	Assemble the final prompt; per-action generation decoration; fastpath context.

Personal-memory surfaces

Module	LOC	Role
personal_memory_surface.py	452	"Is this a personal-memory query" + surface builder.
personal_memory_clean_response.py	338	Clean "what do you know about me" report (reset-aware, poison-filtered, dynamic-fact aging).
personal_memory_deep_response.py	450	Deep memory-internals explain (schema/tables/functions) + routing-fault explain.
profile_extractor.py	1517	Extracts user facts from turns (role/interests/field/"remember that I..."), writes user_patterns + LLM session summaries; recency refresh.
identity_validation.py	176	Validate identity candidates.

Typed pipeline plumbing (evidence/packets)

Module	LOC	Role
evidence_ledger.py	679	Records artifacts/events with signatures; recent generated artifacts; status evidence.
evidence_arbitration.py	212	Scores competing evidence (stage packets + tool results + goals), dedup-by-fingerprint, keep-max.

Module	LOC	Role
stage_packet_store.py, stage_packets.py, pipeline_models.py	~177	The typed packet substrate: route/plan/evidence/generation/output packets flow between stages; this is the plumbing behind “no fake actions”.
background_tasks.py	252	In-process multi-threaded task manager (heavy work → job id → CHECK_JOB).
runtime_policy.py	89	Per-turn budgets/timeouts/context size from runtime snapshot.

Security

Module	LOC	Role
security.py	210	SecurityManager — fail-closed command/path/app allowlist sandbox.

Part 3 — cognition/ module catalogue (47 files, 20.8k lines)

The thinking layer: agents, orchestration, inference, persona, reasoning, governance.

Orchestration & agents

Module	LOC	Role
agent_bus.py	3517	15 specialist agents on a dependency DAG (topological layers) + calibrated weight-free confidence aggregation + per-action agent selection.
orchestrator.py	1123	Gradient 12-stage pipeline (all CHAT modes): planner → retrieve_for_turn() → dispatch_specialists() → heuristic rerank → context assembly. Composes the specialist bus; no longer Quick-only bypass.
learning_coordinator.py	82	Stage 12 finalize_turn() — store assistant turn, publish meta, _learn_from_result().
hyde.py	69	Hypothetical-document- embedding query expansion.
reranker.py	178	Candidate reranking (token overlap + source priority).

Module	LOC	Role
introspection_agent.py	164	Wraps introspection for the bus (pipeline/memory/runtime/audit).
llm_intent.py	260	LLM intent parsing fallback (GGUF, cached).

Inference

Module	LOC	Role
gguf_inference.py	3266	Model-agnostic GGUF inference: model resolution (no baked model), family-aware chat templating, graceful GPU-layer fallback, streaming, output cleaning, token budgeting.
inference_broker.py	222	Thin GGUF broker (infer) used by agents/coding/patching.

Reasoning & engagement

Module	LOC	Role
reasoning_modes.py	671	The 5 modes — canonicalisation, per-mode private system instruction, execution contract (samples/branches/stages + dynamic token budget), reasoning-leak stripping.
engagement_tracker.py	249	Session depth tracking → auto-escalates reasoning mode (Quick→Normal→Advanced→Research) as a conversation deepens; session narrative.
working_memory.py	387	Turn-scoped pinned facts (pin/absorb/evict/persist/restore).

Context & grounding

Module	LOC	Role
context_synthesiser.py	776	Builds the precise prompt context: persona handoff, turns block, vector block, live-runtime brief, budgeting.
grounded_status.py	648	Identity + memory-inventory rendered directly from profile/DBs (direct grounded answers).

Persona (the living voice)

Module	LOC	Role
persona.py	375	Canonical persona authority — base + auto sections, preferences, compose/refresh.
persona_updater.py	748	Re-derives the persona overlay from memory/reflection/habits/runtime patterns; KG population; stale-fact aging.
persona_hygiene.py	130	Cleans/dedups/prunes the auto-persona.
persona_status.py, persona_values.py	~108	Persona status report; values store.

Output governance (consolidated this session)

Module	LOC	Role
output_governor.py	1694	Canonical governance: sanitize, role-prefix/identity-drift repair, self→user confusion repair, style cleanup, confabulation detection, quality scoring, memory-worthiness, GGUF-artifact cleaning (clean_gguf_artifacts), evidence validation.
response_governance.py, response_sanitizer.py	28+14	Re-export shims → output_governor (kept for back-compat).
tone_analyzer.py	371	Analyses recent user turns → tone preferences ELI adapts to over time.

Profile

Module	LOC	Role
user_info_builder.py	752	The living, versioned user profile: multi-source gather, noise filter, categorise, hash, diff-on-change.

Part 4 — remaining packages (module catalogue)

kernel/ (the engine + boot) — 17.5k lines

Module	LOC	Role
engine.py	16014	CognitiveEngine — the conductor: persona, generation settings, the 5 _run_* reasoning passes, grounding overrides, synthesis prompt build + the context-bloat cap, fragment/placeholder guards, dispatch gate to bus vs orchestrator, startup loops (reflection/habit/scheduler/self-improve/proactive).
world_model.py	278	Symbolic self-model: Identity/Runtime/Memory/Goal/Capability states + snapshot/merge.
pipeline_trace.py	85	Canonical S01-S12 stage names and log_pipeline_stage() observability.
state.py	371	User/runtime state + profile (active user id, name, profile text).
self_upgrade.py	575	Self-upgrade orchestrator (git pull, pip, rebuild FAISS/KG, manifest, system index).
scheduler.py	74	Kernel thread-pool/timer scheduler (generic).
pipeline.py	148	Pipeline step description.

core/ (paths, settings, hardware, safety) — 11.2k lines

Module	LOC	Role
runtime_settings.py	1290	Canonical settings.json load/save, legacy-key migration, portable-path healing/sanitising.
hardware_profile.py	2818	Auto-detect GPU/VRAM/RAM → fit ctx/gpu_layers/batch (KV-cache + compute-buffer reserve); model discovery; hardware authority enforcement.
paths.py	768	Single-source-of-truth path resolution (data/config/cache/db/models/voices...).
startup_hardware_optimizer.py	877	Boot-time hardware optimiser (GPU select, layer/ctx allocation, mode presets).
config.py	357	Thin config shim over runtime_settings (canonical key mapping).
model_download.py	673	Curated GGUF catalogue + VRAM-based recommend + download.

Module	LOC	Role
dag.py	470	Generic DAG engine (Kahn topo-order, parallel layers, critical path) — shared by the agent bus + coding engine. Per-boot runtime budget derivation. Offline-by-default socket-level network gate (fail-closed) + guarded_urlopen/http_get_json. Factory-reset of memory/identity (scrub names, clear DBs, backup). User-tunable knowledge-gathering limits + synthesis cap registry (GUI-surfaced). is_grounded_query classifier. STT-robust self-harm detector + persona steering directive. path helpers, ownership map.
dynamic_runtime_budget.py	271	
netguard.py	586	
memory_reset.py	333	
cognition_tunables.py	273	
grounding.py	145	
crisis_guard.py	113	
portable_paths.py, db_paths.py, legacy_paths.py, architecture_contracts.py	small	

memory/ — 8.8k lines

Module	LOC	Role
memory.py	5677	The Memory class (69 public methods): semantic store/recall, conversations, habits, failures/improvements, storage policy, upkeep, weight decay, KG bridge, mark_failure_resolved, disable_invalid_habit_rules. Entity/relation graph + multi-hop BFS (related) + context_for_prompt + extract-from-memory. Habit rules/events store + cheap embed/recall. Shared turn retrieval (retrieve_for_turn) with an 8 s cache, time window and memory_diag stats. Storage policy: origin, dedupe key, forgetting curve, archive rule. Bitemporal claims about the user: valid interval, learned-at, supersession, extraction. Orchestrator stages consume retrieve_for_turn through it.
knowledge_graph.py	643	
habits_memory_db.py	466	
retrieval.py	253	
policy.py	118	
claims.py	266	
unified_retrieval.py	168	

Module	LOC	Role
vector_store.py	637	FAISS index (L2, 1/(1+dist) sim) + nomic embedder + keyword fallback + auto-rebuild.
system_index.py	278	OS app/exe/dir index (the launcher backing — your machine's executables, thousands, indexed live per machine).
memory_truth.py, memory_adapter.py	small	inspection/compat/session helpers.

perception/ — 10.2k lines

Module	LOC	Role
audio_stt.py	2472	faster-whisper STT + VoiceGate (wake-word, debounce, incomplete-command wait), self-echo suppression, output ducking, per-user voice profile bias.
vision.py	698	Local GGUF VL (Moondream fast / primary), hot-swap with text model, CPU-pinned CLIP, OCR.
tts_router.py	1219	Piper/espeak TTS, voice selection, unspeakable-fragment guard.
os_controller.py	572	Screenshot/volume/keys/mouse/clipboard + gaze_click.
screen_locator.py	497	OCR (tesseract) → locate a named UI element on screen.
gaze_engine.py	358	MediaPipe face-gaze + calibration mapper + One-Euro smoothing → latest_gaze.json @10Hz.
local_whisper_stt.py, ambient_vision.py, ana- lyze_{pdfs,image,mesh,csv}.py, log_rotation.py, voice_worker*.py	~1.2k	Whisper backend; ambient glances; file analysers; log rotation; voice workers.

planning/ (autonomy, habits, proactive) — 4.0k lines

Module	LOC	Role
proactive_daemon.py	1523	Background 10-min loop: pattern/code analysis, habit detect+offer, morning report, error tracking.

Module	LOC	Role
autonomy_scheduler.py	334	Policy-gated (observe/proposal/goal-driven) goal scheduler w/ cooldown + attention queue.
habits.py	370	Habit detection (circular time-of-day clustering, shift and lapse notes), offer/pending state, disabled-by-default.
routine_stats.py	125	Circular clustering, routine shift and lapse detection.
habits_scheduler.py	170	Fires active habits at their time (self-heals legacy rows, once-per-minute dedupe).
goal_store.py, goal_models.py, goal_tick.py, operator_goal_actions.py	~400	Mission goals (priority/cadence/risk/constraints/success- criteria) → governed proposals.
attention_queue.py, proposal_queue.py, proposal_*.py, jobqueue*.py, autonomy_controller.py	~900	Attention ranking; proposal queue/archive; job queue; safe autonomy ticks.

coding/ (the frontier coder) — 2.1k lines

Module	LOC	Role
agent.py	215	CodeAgent.solve — composes the loop; broker-backed generator; full provenance.
bug_memory.py	340	Bug classification + (signature→fix) long-term SQLite memory w/ fuzzy recall.
verification.py	353	Gate ladder (syntax→exec→tests) + weighted scoring + test synthesis.
sandbox.py	234	Bounded isolated execution (scrubbed env, CPU limit, timeout).
search.py	176	UCB1 tree search over a temperature-ladder beam.
plan_graph.py	164	Subtask DAG decompose + topological solve + compose.
planner.py, cost.py	~210	Planner/implementer split; cost → foreground/background decision.

learning/ (real LoRA self-training) — 4.3k lines

Module	LOC	Role
lora_trainer.py	834	Real torch/PEFT/transformers Trainer LoRA run.

Module	LOC	Role
<code>lora_trainer_guard.py</code>	630	TrainerTarget plans + safety validation.
<code>dataset_builder.py</code>	552	Build supervised dataset from logged turns/corrections (PII-redacted, deduped).
<code>lora_eval.py</code>	485	Adapter eval suite.
<code>bootstrap_phi3_base.py</code> , <code>base_model_resolver.py</code> , <code>dataset_filters.py</code> , <code>export_trainable_dataset.py</code> , <code>merge_reviewed_datasets.py</code> , <code>training_preflight.py</code>	~1.4k	Trainable base download/resolve, quality gates, export/merge, preflight.

contracts/, integrations/, system/, utils/, cli/

Module	LOC	Role
<code>contracts/runtime_status.py</code>	571	Canonical runtime-status evidence contract (detect question, build live evidence, validate/repair).
<code>contracts/grounded_control.py</code>	216	Grounded-control synthesis guard (evidence-complete? suppress bad clarifications).
<code>integrations/mpris/playerctl_backend.py</code>	751	MPRIS2/playerctl media backend (player resolve, transport, status).
<code>integrations/ollama/client.py</code>	322	Optional Ollama backend client.
<code>system/portable_app_control.py</code>	742	Cross-platform installed-app discovery + open/close/minimise.
<code>utils/platform_compat.py</code>	1666	Cross-platform layer (open url/file/app, volume, clipboard, notify, executables).
<code>utils/log.py</code> , <code>cli/headless.py</code>	~200	Central logger; headless terminal REPL.

world/ (embodied self-model) — 1.8k lines

Module	LOC	Role
<code>agency/autonomy_engine.py</code>	260	Awareness vector (easing/decay) → avatar-room routing + symbolic-object creation during reasoning; provenance/snapshots/journal.
<code>agency/{policy,goal_ecology,habit_manager,reflection_bridge,world_permission_classes.py}</code>	~1.8k	Policy classes; goal decay; default habits; reflection→world; world identity.
<code>core/schemas.py</code> , <code>core/ontology.py</code>	~160	World state schemas (rooms/objects/events/actions/awareness); object templates.

Module	LOC	Role
renderers/pyside6/{world_scene,world_panel}.py	4105	The PySide6 “Eli’s World” tab (house/rooms/avatar/objects, live state).
avatar/*, persistence/*, world_event_bus.py, local_world_bridge.py	~700	Avatar behaviour/locomotion/persona- map; jour- nal/provenance/snapshots/storage; event bus + bridge.

gui/ (PySide6 desktop) — 27.4k lines

Module	LOC	Role
eli_pro_audio_gui_v2_0.py	13149	Main window: 13 top-level tabs + adapters (CentralMemory/LocalModel/Ollama/Executor bridges, the _GUIEngineAdapter), chat, drag-drop, reasoning-mode auto-select, all toggles.
labs_tab.py	5744	Labs workspace: Notebook, Memory browser, Jupyter launcher, Calculator(+constants), Physics tables, Report Builder (evidence-grounded docs), File-Chat, Workspaces, Sim-IDE.
app.py	827	Launcher / first-boot auto-tune / main().
panels/startup.py	1974	StartupModelSelectionDialog, FirstBootWizard, HardwareTuningDock.
panels/settings.py	825	Advanced Settings (Agents/Models/ Cognition /Plugins/Self-Upgrade).
docks/operator_console_dock.py	303	Governed operator console (proposals/goals/policy/attention).
widgets/, tabs/, panels/agent_wizard.py, docks/proactive_dock.py, qt_compat.py	~1k	Ollama selector, experimental/world tabs, agent wizard, proactive dock, Qt shim.

plugins/ — 10 built-ins + manager

Module	Role
manager.py (553L)	Discover/install/enable/disable/execute; auto-load; builtin-stub gen; registry.
web, web_automation, weather, calendar, notes, pomodoro, system_stats, media, tts, document_reader	The 10 built-in plugins (see Part 1 §17).
base/base.py	Plugin base class + action validation + loader.

Module	Role
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Part 7 — modules not covered above

Every remaining module under `eli/`, with its line count and a one-line role (the module's own docstring where it has one). Regenerate the counts with the snippet in `tools/refresh_doc_metrics.py`.

`runtime/`

Module	Lines	Role
<code>active_project.py</code>	78	Active project signal.
<code>api_users.py</code>	145	API users + roles (admin / member RBAC) for the web server.
<code>auth.py</code>	4	Tombstone comment: the unfinished multi-user AuthManager was removed; authority lives in <code>authority_gate.py</code> , <code>security.py</code> and the engine's persona lock.
<code>authority_gate.py</code>	49	<code>authority_gate</code> — intentional stub.
<code>autopilot_debugger.py</code>	377	Autopilot debugger — one loop that turns a failure into a plan.
<code>ble_light.py</code>	247	BLE light control — the GATT writes ELI never had.
<code>bt_platform.py</code>	1335	Cross-platform Bluetooth radio detection, recovery, and classic discovery.
<code>codebase_graph.py</code>	306	Codebase self-graph — ELI's live, grounded model of how its OWN code connects.
<code>command_splitter.py</code>	84	Split one utterance that chains MULTIPLE imperative commands into its parts.
<code>conversation_thread.py</code>	284	Conversation thread awareness — topic carryover, proactive grounding, web queries.
<code>desktop_launchers.py</code>	718	Redistribution-safe desktop / Start Menu launchers.
<code>deterministic_failure_patches.py</code>	108	Rule-based code patches for recurring executor failures without LLM guessing.
<code>device_drivers.py</code>	1124	Pluggable LOCAL-control drivers for ELI's device server — control devices that don't

Module	Lines	Role
device_names.py	219	User-chosen device names — stable keys for voice control across hardware/OS.
device_server.py	1453	ELI's own device server — original, MQTT-first. No Home Assistant.
eval_review.py	117	Eval + LLM-judge board, in the shape Labs ▶ Test & Review already renders.
evidence_planner.py	302	Evidence planner + gatherer — the DAG/plan principle for generative & grounded tasks.
failure_taxonomy.py	230	Classify a runtime failure by what actually went wrong, and where.
gguf_runtime_report.py	257	Portable GGUF / inference diagnostics — model-agnostic, path-agnostic.
home_intel.py	114	ELI home intelligence — weave the LLM into the home/device system.
home_mesh.py	435	ELI home mesh — tiered brains with LAN failover.
inference_footprint.py	413	Live inference RAM — read from the loaded llama.cpp model and this process, never guessed.
last_trace.py	58	Persists the last response trace, keyed to the process id so a restart never reports a closed session's turn.
license_info.py	99	Locate and print ELI's licence, wherever ELI happens to be running from.
local_connectivity.py	764	Local-only WiFi and audio routing — sovereign stack, no cloud.
memory_provenance.py	153	Memory provenance and verification tier for grounded recall.
mqtt_setup.py	444	Cross-platform MQTT broker onboarding for ELI redistribution.
native_locks.py	14	Process-wide locks for native llama.cpp and FAISS entry points, preventing cross-context heap corruption.
proposal_adapters.py	1	Re-export shim of planning/proposal_adapters.py.

Module	Lines	Role
reflection.py	377	Reflection engine — analyses memories, conversations, and patterns to extract insights.
lessons.py	345	Expiring, checked lessons from recurring failures (trigger, evidence, predicted outcome, retire when unhelpful).
relational_facts.py	83	Extract relational facts the user mentions in passing.
repair_playbook.py	243	ELI repair playbook — decision guide for self-maintenance, code examine, and upgrades.
report_pipeline.py	198	Multi-stage grounded document pipeline (Report-Builder discipline, chat scale).
research_corpus.py	439	Local research corpus workspaces.
route_authority.py	1	Re-export shim of execution/route_authority.py.
scheduled_tasks.py	569	Scheduled / overnight advanced tasks.
self_facts.py	333	Verified facts about ELI's own construction, assembled from live sources.
self_maintenance.py	175	Unified self-maintenance orchestration for ELI.
self_maintenance_config.py	6	Shared constants for ELI self-maintenance (analysis windows, cluster gates).
self_model_refresh.py	61	refresh_all_overlays_nonfatal: refreshes the persona and user-profile overlays without raising.
self_status.py	264	Real, measured self-status for ELI.
server_util.py	240	Helpers for detecting an existing ELI web server and firewall guidance.
session_continuity.py	100	Session continuity — live thread memory injected into every CHAT turn.
shell_followup.py	68	Resolve deictic shell follow-ups (“run that command”) from prior assistant output.
state_providers.py	80	State providers — the hook that lets components expose save/restore state to a
test_generator.py	283	ELI-assisted behavioural test generation (Phase 4).

Module	Lines	Role
test_review.py	234	Full test/project run → review workflow.
user_model.py	372	Continuous User Model — ELI's living, semantic, auto-updating model of the user.
voice_assets.py	558	Ensure the local voice models (STT + TTS weights) are present.

cognition/

Module	Lines	Role
agent_spec.py	436	A real specification for a custom agent — objective, prompt, triggers, measures.
agent_trust.py	262	Trust for custom agent code — provenance, not just a hash in a dict.
belief.py	239	Belief revision — what ELI holds, how strongly, and what it takes to change it.
chat_grounding_gate.py	157	Fail-closed CHAT gate — skip LLM when user-fact grounding is required but absent.
context_budget.py	128	Sizing and trimming the memory block that goes into a prompt.
correction_patterns.py	224	Shared patterns for user correction / dispute turns.
emotion_palette.py	254	Emotion / tone palette — the shared taxonomy ELI expresses through.
emotion_timeline.py	452	Emotion timeline — the durable record of how the USER has been feeling.
evidence_format.py	52	Dates on evidence lines, from each row's own timestamp.
expression_state.py	63	Live avatar/expression state — the thin bridge that lets ELI's face react in
memory_diag.py	73	What memory retrieval actually did this turn, from telemetry, so ELI never has to guess why.
model_identity.py	237	Model-agnostic identity: chat family + thinking from GGUF metadata.
model_load_diagnostics.py	633	Say why a GGUF would not load — for ANY model, not a known list of them.

Module	Lines	Role
model_output_tokens.py	301	Canonical special tokens, stop sequences, and persona drift patterns for all GGUF families.
personal_context_gate.py	118	Gate when past-session plans/projects may enter the persona handoff.
query_planner.py	161	Deterministic query planning: the time window a question asks about.
scoring.py	116	Canonical scoring + confidence primitives for ELI.
stance_capture.py	151	Notice when ELI has taken a position, so it survives the conversation.
stance_store.py	288	Persistence for beliefs — ELI's own positions, and the record of revisions.
tone_adaptor.py	224	Tone adaptor — decides the emotion ELI <i>expresses</i> and feeds every output channel.
turn_dossier.py	404	Turn dossier — single assembly point for awareness, memory, and cognition context.
user_claim_validator.py	258	Post-generation validation of user-attribution claims in CHAT output.

execution/

Module	Lines	Role
app_aliases.py	90	Speech-damage layer for app names, on top of the cross-platform resolver.
effectors/system_helpers.py	175	Cross-platform system helpers shared by executor and effectors (v3).
execution_planner.py	111	Builds RouteDecision and the typed ExecutionPlan/PlanStep artifacts the bus consumes.
media_runtime.py	709	Media runtime control for ELI.
operator_actions.py	105	Valid states and helpers for operator (proposal) actions: pending, approved, rejected, blocked, pending_confirmation.

Module	Lines	Role
operator_policy.py	65	The valid operator policy modes: proposal_only, operator_supervised, goal_driven, observe_only.
portable_intent_contract.py	643	Portable intent rules; hard-blocks document, code and analysis prompts from PLAY_MEDIA.
route_authority.py	46	Thread-local record of which layer decided a route, and the internal prompt prefixes that must never be routed as user text.
route_contracts.py	123	Small predicates that classify a request (for example wants_memory_internals) for the router.
router_enhanced.py	8343	The router: route(text) regex-first priority pipeline with LLM-intent fallback (see architecture.md §4).
shell_gate.py	110	Centralised shell-command safety gate.

core/

Module	Lines	Role
confidence.py	9	The one confidence-score to label mapping.
full_control.py	46	ELI Full Control — a single master override that lifts ELI's safety barriers.
gpu_pack_runtime.py	83	Activate the optional GPU pack before llama_cpp is imported (AppImage/portable).
init_data.py	283	First-run data initialiser — build ELI's FULL database architecture up front.
llama_cpu_compat.py	240	CPU compatibility for llama-cpp-python — parity with GPU probe/fallback policy.
load_probe.py	479	Verify a set of load parameters actually works, without dying if it doesn't.
model_tier.py	141	Model-capability tier — the single signal that lets ELI's cognition budgets
secure_io.py	94	Race-free writing of files that must never be world-readable.

Module	Lines	Role
self_provenance.py	207	Is this row ELI's own bookkeeping, or evidence about the user?
sqlite_util.py	99	Shared SQLite pragmas — WAL with a portable-safe fallback.
toml_util.py	19	Load TOML on Python 3.10+ (tomli backport) and 3.11+ (stdlib tomllib).
worldclock.py	348	Place-aware wall-clock answers.

memory/

Module	Lines	Role
unified_retrieval.py	168	Unified memory retrieval for orchestrator and agent bus.

perception/

Module	Lines	Role
desktop_capabilities.py	186	Cross-OS desktop control capability probe — one place for per-platform truth.
mic_resolver.py	613	ELI microphone resolver — pick a capture device that actually delivers audio,
screen_analysis.py	194	Local screen analysis — depth modes, memory recall, research context.
tts_xtts.py	394	Voice cloning backend (Coqui XTTS-v2).
ui_ground.py	322	Optional UI grounding backends — 100% local by default.
ui_tree.py	353	Accessibility-tree UI targeting (Linux / AT-SPI).
voice_fx.py	252	Character-voice effects layer.
voice_profile.py	417	Voice profile + prosody — the foundation for tone/emotion detection.
voice_worker_streaming.py	61	Stream-aware voice worker for ELI.
wakeword.py	456	Self-trained, fully-local wake-word detector (openWakeWord features + a custom head).

planning/

Module	Lines	Role
goal_autogenesis.py	224	eli.planning.goal_autogenesis
insight_synthesis.py	135	Cached, background-synthesised reflection insight.
jobqueue.py	184	Durable SQLite-backed job queue for external subprocess jobs; files under <data>/jobs.
jobqueue_cli.py	59	Durable background jobs that run as separate processes and survive a restart.
proposal_adapters.py	165	Adapters that make proposal execution results look like subprocess.CompletedProcess.
proposal_memory_bridge.py	64	Bridge that records proposal outcomes into memory through safe calls.
proposal_models.py	48	ProposalRecord: kind, payload, source, priority.

coding/

Module	Lines	Role
code_mode.py	91	Code-mode — the model writes a small program against ELI's own capabilities.
repo_context.py	91	Repo-context retrieval for the coding agent (Advancement C).
restricted_exec.py	170	In-process restricted execution for code-mode.

learning/

Module	Lines	Role
lora_pipeline.py	149	LoRA training pipeline — the DAG that chains the standalone learning stages.
review_queue.py	270	The human review gate that had no way to be passed.
target_registry.py	266	User-declared LoRA training targets — the registry that replaces the Phi-3 lock.

integrations/

Module	Lines	Role
local_gguf/inference.py	2	Compatibility shim: canonical GGUF inference lives in
mcp/server.py	194	eli.cognition.gguf_inference. MCP server — ELI's own capabilities exposed to any MCP client.
media/capabilities.py	181	Runtime media capability probe — what works on THIS machine/OS.
media/cross_platform.py	590	Cross-platform Spotify control and mpv IPC.
media/media_deps.py	141	Discover media CLI tools — bundled install root, then system PATH.
media/spotify_intent.py	139	Parse user phrasing into Spotify playback intents (no Web API).
media/youtube_playback.py	339	Shared YouTube playback helpers — mpv, yt-dlp clients, browser URLs.

system/

Module	Lines	Role
process_guard.py	447	Verified process-kill safety for CLOSE_APP and friends.

utils/

Module	Lines	Role
jsonio.py	40	Small tolerant JSON readers shared across the runtime.
native_io.py	74	Silence stderr written by C extensions, at the file-descriptor level.

onboarding/

Module	Lines	Role
interview.py	701	Light, skippable first-run onboarding interview.

setup/

Module	Lines	Role
<code>__main__.py</code>	48	Entry: <code>python -m eli.setup</code> [-wizard] [-launch] [-status] [-full-install]
<code>hardware_policy.py</code>	330	Install-time accelerator inventory and CPU-only recommendation.
<code>install_backend.py</code>	185	Cross-platform install backend — streams <code>install.sh</code> / <code>install.ps1</code> into GUI progress.
<code>install_messages.py</code>	116	Curated installer copy — ELI voice: direct, dry, nerdy, occasionally a pisstaker.
<code>platform_profile.py</code>	142	Install-time platform detection — routes GUI, headless, and Android profiles.
<code>status.py</code>	120	Read-only setup completeness checks for first-run / grandparent setup.
<code>unified_installer.py</code>	686	One-click GUI installer — backend terminal work with OS-style progress and ELI wit.
<code>wizard.py</code>	298	Grandparent-ready graphical setup wizard — every first-run stage in one place.

world/

Module	Lines	Role
<code>avatar/behaviour_controller.py</code>	36	<code>AvatarBehaviourController.choose_room</code> : picks the room from the awareness vector (anomaly room on low evidence, memory archive on low memory confidence).
<code>avatar/locomotion.py</code>	17	Room coordinates and <code>target_for_room</code> .
<code>avatar/persona_mapper.py</code>	40	<code>PersonaToAvatarMapper</code> : maps persona and awareness to avatar tint and expression.
<code>persistence/journal.py</code>	25	World journal path and append helper (<code>journal_path</code> , <code>append_journal_entry</code>).
<code>persistence/provenance.py</code>	22	World provenance ledger (<code>ledger_path</code> , provenance ids).
<code>persistence/snapshots.py</code>	35	World snapshots (<code>snapshot_dir</code> , <code>create_snapshot</code>).

Module	Lines	Role
persistence/storage.py	267	World state directory resolution and load and save of world state.

gui/

Module	Lines	Role
branding.py	143	Shared ELI branding assets (window icon, Freedesktop theme, Windows .ico).
code_editor.py	220	Native code editor for the IDE — PySide6/PyQt, no QScintilla required.
coding_tab.py	178	ELI v2.0 — Coding tab.
panels/_qt.py	33	Shared Qt import shim for eli.gui.panels modules.
panels/permission_dialog.py	232	The consent dialog — the moment the operator actually decides.
tabs/eli_world_tab.py	11	The Eli's World tab: wraps EliWorldPanel.
tabs/experimental_tab.py	170	Read-only workbench over the experimental/ folder: lists prototype kits and opens their folders or READMEs.
tabs/marketplace_tab.py	991	Settings ► Plugins ► Marketplace — browse, scan, install, and revoke.
tabs/tasks_tab.py	284	ELI v2.0 — Tasks tab.
tabs/training_tab.py	841	ELI v2.0 — Labs ► Training. The GUI half of the LoRA pipeline.
widgets/eli_face.py	230	ELI's animated face — a procedural 2D face that shows the tone_adaptor's emotion.
widgets/ollama_model_selector.py	444	Ollama model selector widget for ELI GUI.
widgets/voice_downloader.py	282	Voice library browser — acquire additional Piper voices, accents and tones.

plugins/

Module	Lines	Role
calendar/plugin.py	109	Calendar plugin: ADD_EVENT and LIST_EVENTS over ICS events.

Module	Lines	Role
document_reader/plugin.py	355	Document reader plugin: reads and indexes PDF and docx files.
integrity.py	291	Integrity and publisher identity for community plugins.
manifest.py	302	Plugin manifests: what a plugin says it is, checked against what it does.
marketplace.py	793	ELI's plugin marketplace client — federated, community-hosted, consent-gated.
mcp.py	481	MCP servers: one config ELI owns, and an install that proves it works.
media/plugin.py	93	Media control plugin for ELI.
notes/plugin.py	87	Notes plugin: markdown notes with NEW_NOTE, WRITE_NOTE, LIST_NOTES, SEARCH_NOTES.
permissions.py	395	Android-style consent for plugins: allow always, allow once, reject.
pomodoro/plugin.py	92	Pomodoro plugin: POMODORO_START, POMODORO_STOP, POMODORO_STATUS focus timers.
security_scan.py	547	Multi-engine malware scanner for community plugins.
subprocess_sandbox.py	308	Containment for child processes — the gap netguard structurally cannot cover.
system_stats/plugin.py	74	System stats plugin: CPU_USAGE, RAM_USAGE, SYSTEM_STATS.
tts/plugin.py	43	TTS (Text-to-Speech) plugin for ELI.
weather/plugin.py	132	Weather plugin: GET_WEATHER (geocode plus open-meteo, toggle-gated).
web/plugin.py	617	Web plugin: WEB_SEARCH (toggle-gated).
web_automation/plugin.py	92	Web Automation plugin for ELI – lazy Playwright import, safe under broken installs.

tools/

Module	Lines	Role
api.py	73	eli.api — a small, curated callable surface over ELI's action executor.
image_engine/fetch_model.py	59	Fetch a diffusers image model's weights into the local models dir.
image_engine/gui_bridge.py	453	Bridge between the GUI image tab and the engine: ImageGenerationRequest, model and preset discovery, recent outputs.
image_engine/image_engine/__main__.py	3	python -m entry for the image engine CLI.
image_engine/image_engine/cli.py	312	Command-line interface for the image engine.
image_engine/image_engine/contracts.py	126	EngineConfig, JobRequest, ArtifactRecord, JobResult.
image_engine/image_engine/plotting.py	324	Loads records and plots them with matplotlib.
image_engine/image_engine/project_analyzer.py	41	analyze_project_profile, infer_style_hints: derives a colour and style profile from a project path.
image_engine/image_engine/prompt_compiler.py	118	VisualBrief and PromptCompiler: turns a request into a structured visual brief.
image_engine/image_engine/quality.py	94	Image hashing and scoring (score_image, choose_best).
image_engine/image_engine/service.py	369	ImageEngine: runs image jobs and records artifacts.
image_engine/image_engine/visual_core.py	1446	The procedural renderer: scene types, composition, palettes, atmosphere and post-processing.
image_engine/runtime_paths.py	130	Resolves image-engine paths from the project root or the data directory.
mic_diag.py	149	ELI Microphone Diagnostic
news/news_fetcher.py	577	ELI Web Learning Module — News & Current Events Fetcher
news/news_synthesis.py	913	News synthesis cadence
registry/capabilities.py	142	Capabilities management for ELI.
registry/capabilities_doc.py	374	capabilities_doc.py — regenerate blueprints/capabilities_and_actions.md.
registry/capability_registry.py	38	Process-local capability registry (register_capability).
registry/capability_updater.py	142	capability_updater.py

package root

Module	Lines	Role
<code>__main__.py</code>	93	Console entry point. <code>python -m eli</code> and the <code>eli</code> script both call this.

scripts/

Module	Lines	Role
<code>rebuild_vector_index.py</code>	141	<code>rebuild_vector_index.py</code>

Part 5 — engine request lifecycle (`engine.process`)

The entry flow, in order:

1. **Pipeline trace id** assigned (`ELI_PIPELINE_TRACE` for observability).
2. **Prompt-injection guard FIRST** — `_eli_sanitize_user_input` runs before the router or LLM ever sees the raw text, so injected instructions can't reach them unsanitised. (A security control that sits at the very top of the pipe.)
3. **Multi-question splitter** (`kernel/engine.py`) — if a message has more than one `?`, more than 25 words and is not conversational, it splits into standalone sub-questions (each at least 6 words, at most 4) and answers each, so genuinely multi-part asks get multiple grounded answers.
4. Router → **gradient orchestrator** (all CHAT modes, depth by reasoning mode) → shared retrieval → specialist bus composition → governed output (Parts 1–4). Fallback: `direct AgentBus.dispatch()` if orchestration fails.

Note: `engine.py` still carries 34 PHASE... markers: layered patches whose residue (engine 16.0k / executor 15.8k / GUI 13.1k lines) is documented here at behavioural level, not line by line.

Part 6 — reasoning modes & autonomous deepening

The 5 reasoning modes were renamed (public layer; internal keys unchanged) and made adaptive + self-deepening.

Public mode	Internal key	Multi-pass algorithm	Agent time budget	Deepen iters	Confidence target
Quick	<code>quick</code>	single pass	1.0×	0 (never)	0.45
Normal	<code>chain_of_thought</code>	one reasoned pass	1.0×	1	0.55
Advanced	<code>self_consistency</code>	N samples + consensus	1.5×	2	0.65
Research	<code>tree_of_thoughts</code>	branch + prune	2.0×	3	0.75
Expert	<code>constitutional_ai</code>	draft → critique → revise	2.5×	4	0.80

Auto-selection (cognition/engagement_tracker.py): the mode is escalated automatically as a conversation deepens (Quick→Normal→Advanced→Research) without the user asking. (Internal strategy keys are stable; users only ever see the public names.)

Per-mode agent time budgets (reasoning_modes.mode_budget_multiplier, Stage 1b): each mode scales every agent's timeout — quick/normal unchanged, deeper modes get proportionally more time. Threaded via intent["_mode_budget_mult"] → agent_bus._collect_layer. Tunable in the Cognition tab (cog.mode_budget_*).

Confidence-driven iterative deepening (runtime/grounding_escalation.escalate, Stage 2): on a poorly-grounded *checkable-factual* turn, ELI re-dispatches the agent bus broadly, **escalating the reasoning mode one tier per iteration** and (Stage 3a) **raising the gather budget** each pass ($1.5\times\rightarrow2.0\times\rightarrow2.5\times$), until grounding crosses the per-mode target or the per-mode iteration budget is spent (early no-improvement break). Banter/opinion/command/meta never deepen.

Background deepening (runtime/background_deepening.py, Stage 3b): a Quick answer returns instantly; if it was a poorly-grounded factual turn, a background thread keeps gathering (broad re-dispatch at Advanced + $2\times$ gather) and surfaces a better answer in the Proactive panel **only if** it crosses grounding 0.55 and isn't degenerate. Tightly gated (quick-only, grounding<0.40, factual-only, deduped, 1 in-flight, 10-min cooldown). Toggle: cog.background_deepen / ELI_BACKGROUND_DEEPEN.